A Lean-verified offset construction for the Erdős–Straus equation

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Abstract

We give a self-contained formulation of an offset construction for the Erdős–Straus equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

The construction packages the classical divisor split into a congruence condition indexed by a prime $p \equiv 3 \pmod 4$. Its full positive-integer statement is formalised in Lean 4 with Mathlib: the development constructs the denominators by natural-number division, proves the divisions exact and the denominators positive, and concludes the equality in $\mathbb{Q}$. We also formalise a fixed-divisor lifting theorem, prove that $4pd$ is its least positive transport period, and give three groups of explicit infinite families, including the progressions arising from $p = 3$ and $d_1 = 2, 5$. We make no claim toward resolving the Erdős–Straus conjecture; the underlying divisor split is classical.

1 The divisor split

The Erdős–Straus conjecture asks whether, for every integer $n \geq 2$, there exist $x, y, z \in \mathbb{N}_{>0}$ satisfying

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

The conjecture remains open. We use the following classical algebraic reduction; for historical background, see Mordell [1] and Vaughan [2].

Fix $x \in \mathbb{N}_{>0}$ with $4x > n$, and set

$$a = 4x - n, \qquad b = nx.$$

Then

$$\frac{4}{n} - \frac{1}{x} = \frac{a}{b},$$

and multiplication and rearrangement give

$$(ay - b)(az - b) = b^2. \tag{2}$$

Lemma 1.1 (positive divisor split)**.** *Let $n, x, a, b, d_1, d_2, y, z \in \mathbb{N}_{>0}$. Suppose*

$$a = 4x - n, \qquad b = nx, \qquad d_1 d_2 = b^2,$$

and

$$ay = b + d_1, \qquad az = b + d_2.$$

Then (x, y, z) satisfies (1).

Proof. The last three equations are precisely the factorisation (2). Expanding it and using $a = 4x - n$ and $b = nx$ gives

$$4xyz = n(xy + xz + yz).$$

All four variables n, x, y, z are nonzero, so division by $nxyz$ gives (1). $\square$

The Lean development separates these two steps. It first proves the cross-multiplied identity over $\mathbb{Z}$, then proves that a positive instance over $\mathbb{N}$ gives the rational equality in $\mathbb{Q}$.

2 A positive certificate

The offset form avoids subtraction and is convenient both mathematically and formally.

Theorem 2.1 (positive offset certificate). *Let $n, x, p, b, d_1, d_2, y, z \in \mathbb{N}$, with $n, x, p, y, z > 0$. If*

$$4x = n + p, \qquad b = nx, \tag{3}$$

$$d_1 d_2 = b^2, \qquad py = b + d_1, \qquad pz = b + d_2, \tag{4}$$

then (x, y, z) satisfies (1).

Proof. The equations are the positive divisor split with $a = p = 4x - n$. Equivalently, substitution in the cross-multiplied target leaves the multiple

$$p\,(d_1 d_2 - b^2),$$

which vanishes by (4). Positivity then permits passage to the rational identity. $\square$

In Lean, `IsES n x y z` contains both the four positivity assertions and the equality (1) in $\mathbb{Q}$. Theorem 2.1 is the declaration `isES_of_offset_certificate`.

3 The constructive offset theorem

Theorem 3.1 (constructive offset theorem). *Let $n, p, d_1 \in \mathbb{N}$ satisfy*

$$n \geq 2, \qquad p \text{ prime}, \qquad p \equiv 3 \pmod 4, \qquad n \equiv 1 \pmod 4, \qquad p \nmid n.$$

Set

$$x = \frac{n+p}{4}, \qquad b = nx.$$

Suppose $d_1 > 0$, $d_1 \mid b$, and

$$p \mid b + d_1, \tag{5}$$

equivalently $d_1 \equiv -b \pmod p$. Define

$$d_2 = \frac{b^2}{d_1}, \qquad y = \frac{b + d_1}{p}, \qquad z = \frac{b + d_2}{p}. \tag{6}$$

Then $x, y, z \in \mathbb{N}_{>0}$, all divisions in (6) are exact, and (x, y, z) satisfies (1).

Proof. The congruences on n and p show that $4 \mid n + p$, so $x \in \mathbb{N}$ and $4x = n + p$. Moreover $x > 0$. If $p \mid x$, then $p \mid 4x = n + p$, hence $p \mid n$, a contradiction. Thus $p \nmid b = nx$.

Since $d_1 \mid b$, the quotient $d_2 = b^2/d_1$ is exact and positive. It remains to prove $p \mid b + d_2$. Work in the field $\mathbb{Z}/p\mathbb{Z}$. Condition (5) gives

$$d_1 = -b.$$

The element b is nonzero, hence so is d_1. From $d_1 d_2 = b^2$, we may cancel d_1 to obtain

$$d_2 = -b,$$

and therefore $p \mid b + d_2$. Both definitions of y, z in (6) are consequently exact. Their numerators are positive, so $y, z > 0$. Theorem 2.1 now applies. $\square$

Remark. The primality assumption is used only for cancellation modulo p. The Lean proof implements this step in `ZMod p`; the remainder of the construction uses natural numbers, with the final equality stated over the rationals.

4 Fixed-divisor lifting

For fixed p, define

$$X_p(n) = \frac{n+p}{4}, \qquad B_p(n) = nX_p(n),$$

where division is natural-number division. Call a triple (p, d, n) *offset-admissible* if

$$n \geq 2, \quad p \text{ prime}, \quad p \equiv 3 \pmod 4, \quad n \equiv 1 \pmod 4, \quad p \nmid n,$$
$$d > 0, \quad d \mid B_p(n), \quad p \mid B_p(n) + d.$$

These are exactly the hypotheses of Theorem 3.1, packaged with a fixed divisor d.

Theorem 4.1 (fixed-divisor lifting)**.** *If (p, d, n) is offset-admissible, then for every $k \in \mathbb{N}$,*

$$n_k = n + 4pdk$$

is offset-admissible for the same p, d. Consequently, on setting

$$x_k = X_p(n_k), \qquad b_k = B_p(n_k), \qquad d_{2,k} = \frac{b_k^2}{d},$$

the positive natural numbers

$$y_k = \frac{b_k + d}{p}, \qquad z_k = \frac{b_k + d_{2,k}}{p}$$

satisfy

$$\frac{4}{n_k} = \frac{1}{x_k} + \frac{1}{y_k} + \frac{1}{z_k}.$$

Moreover, $k \mapsto n_k$ is strictly increasing, so these parameters form an infinite arithmetic progression.

Proof. The congruences defining admissibility make $n + p$ divisible by 4. For $n_k = n + 4pdk$, exact division gives

$$X_p(n_k) = X_p(n) + pdk. \tag{7}$$

Expanding $B_p(n_k) = n_k X_p(n_k)$ using (7) gives the exact identity

$$B_p(n_k) = B_p(n) + pdk\big(n + 4X_p(n) + 4pdk\big). \tag{8}$$

The added term in (8) is divisible by both d and p. It follows respectively that $d \mid B_p(n_k)$ and $p \mid B_p(n_k) + d$. Also $n_k \equiv n \pmod 4$ and $n_k \equiv n \pmod p$, so the two modular hypotheses are preserved; all other admissibility conditions are unchanged or immediate from $n_k \geq n$. Theorem 3.1 supplies the displayed positive representation, including exactness of all divisions. Finally,

$$n_{k+1} - n_k = 4pd > 0$$

because a prime p and the admissible divisor d are positive. $\square$

5 Exact and least transport steps

The step $4pd$ in Theorem 4.1 is sufficient, but its proof does not by itself decide whether a smaller step works for a particular admissible seed. The following criterion answers that question.

Theorem 5.1 (transport-step criterion)**.** *Let (p, d, n) be offset-admissible and let $T > 0$. Put $u = T/4$, using natural-number division. Then*

$$(p, d, n + Tk) \text{ is offset-admissible for every } k \in \mathbb{N}$$

if and only if

$$4 \mid T, \qquad p \mid T, \qquad d \mid 8u^2, \qquad d \mid u\big(n + 4X_p(n) + 4u\big). \tag{9}$$

Proof. If the progression remains admissible, its congruence modulo 4, evaluated at $k = 1$, forces $4 \mid T$. Also $p \mid T$: otherwise T is invertible modulo p, and some residue k satisfies $n + Tk \equiv 0 \pmod{p}$, contrary to admissibility.

Write $T = 4u$. Exact expansion gives

$$B_p(n + 4uk) - B_p(n) = u\big(n + 4X_p(n)\big)k + 4u^2k^2. \tag{10}$$

For natural numbers a_1, a_2, d,

$$d \mid a_1k + a_2k^2 \quad \text{for every } k \in \mathbb{N}$$

if and only if

$$d \mid a_1 + a_2, \qquad d \mid 2a_2.$$

The forward implication follows by evaluating at $k = 1, 2$; the reverse implication uses that $k(k - 1)$ is even. Applying this observation to (10) yields the last two conditions in (9).

Conversely, the four conditions preserve respectively the congruence modulo 4, nondivisibility by p, and the two divisibility clauses involving B_p. The remaining positivity and primality clauses are unchanged, so every transported triple is offset-admissible. □

Theorem 5.2 (least transport period)**.** *For every offset-admissible (p, d, n) and every $T > 0$,*

$$(p, d, n + Tk) \text{ is offset-admissible for every } k \in \mathbb{N} \iff 4pd \mid T.$$

Consequently, $4pd$ is the least positive transport period.

Proof. Only necessity remains after Theorem 4.1. Apply Theorem 5.1 and write $T = 4u$. Since $p \mid T$ and $p \nmid 4$, we have $p \mid u$.

We next prove $d \mid u$. Put $C = n + 4X_p(n) = 2n + p$, and let q be any prime divisor of d. Since $d \mid B_p(n) = nX_p(n)$, either $q \mid n$ or $q \mid X_p(n)$. In the first case, $q \mid C = 2n + p$ would imply $q \mid p$. In the second, the identities $4X_p(n) = n + p$ and $C = (n + p) + n$ first imply $q \mid n$, reducing to the first case. Primality would then give $q = p$, contrary to $p \nmid n$. Thus $q \nmid C$.

We claim that $q \nmid C + 4u$. If $q \mid u$, this follows immediately from $q \nmid C$. If $q \nmid u$, the divisibilities $q \mid d$ and $d \mid 8u^2$ force $q \mid 8$, hence $q = 2$. But $n \equiv 1 \pmod{4}$, so $C + 4u$ is odd, again a contradiction.

Let q^e be the full q-power dividing d. From

$$d \mid u(C + 4u)$$

and $q \nmid C + 4u$, prime-power cancellation gives $q^e \mid u$. This holds for every prime q, so $d \mid u$. Finally p and d are coprime: if $p \mid d$, then $p \mid B_p(n) + d$ would imply $p \mid B_p(n)$, contradicting $p \nmid n$ and $p \nmid X_p(n)$. Therefore $pd \mid u$, and hence $4pd \mid T$. □

For example, the theorem proves that the seed $(p, d, n) = (3, 2, 5)$ has least positive transport period 24; in particular, the tempting half-step $T = 12$ fails. This negative check is included as a Lean regression.

6 Explicit infinite families

We record the families formalised in the accompanying Lean file. In every corollary, $k \in \mathbb{N}$.

Corollary 6.1 (Family I). *Let*

$$n = 12k + 9, \qquad x = 3k + 3, \qquad b = n(k + 1).$$

Then

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{b+1} + \frac{1}{b(b+1)}.$$

Proof. Here $4x - n = 3$ and $\gcd(3, nx) = 3$, so the reduced numerator is 1. Taking $d_1 = 1$ and $d_2 = b^2$ in the divisor split gives the displayed denominators. The Lean proof is a direct positive polynomial identity. □

Corollary 6.2 ($p = 3, d_1 = 2$). *Let*

$$n = 24k + 5 \qquad \text{or} \qquad n = 24k + 13.$$

Put $x = (n + 3)/4$ and $b = nx$. Then

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{(b+2)/3} + \frac{1}{(b + b^2/2)/3}.$$

Proof. The triples $(3, 2, 5)$ and $(3, 2, 13)$ are offset-admissible. Applying Theorem 4.1 gives periods $4 \cdot 3 \cdot 2 = 24$ and hence the two displayed progressions. □

Corollary 6.3 ($p = 3, d_1 = 5$). *Let*

$$n \in \{60k + 5,\ 60k + 17,\ 60k + 25,\ 60k + 37\}.$$

Put $x = (n + 3)/4$ and $b = nx$. Then

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{(b+5)/3} + \frac{1}{(b + b^2/5)/3}.$$

Proof. The triples with $p = 3, d_1 = 5$ and seeds

$$n \in \{5, 17, 25, 37\}$$

are offset-admissible. Theorem 4.1 gives period $4 \cdot 3 \cdot 5 = 60$, yielding the four displayed progressions. □

The two progressions in Corollary 6.2 are the nonmultiples of 3 in the classical Mordell region $n \equiv 5 \pmod 8$; the remaining progression $n = 24k + 21$ is contained in Family I. The purpose of including them is to exhibit the general construction and its formal specialisation, not to claim a new residue family.

7 Lean formalisation

The formalisation is contained in `lean/ESTheorem.lean` and uses Lean 4 with Mathlib. The complete source is available at https://github.com/carlok/erdos-straus-offset-lean. Its public positive interface consists of:

- `IsES` and `isES_of_polynomial`;

- `isES_of_offset_certificate`;
- `second_divisibility`;
- `constructive_offset`;
- the reducible definitions `offsetX`, `offsetB`, and the packaged interface `OffsetAdmissible` with `OffsetAdmissible.isES`;
- `offsetX_add_period`, `offsetB_add_period`, and `OffsetAdmissible.add_period`;
- `offsetX_add_general`, `offsetB_add_general`, `poly_dvd_all`, and the exact transport-step criterion `OffsetAdmissible.all_add_mul_iff`;
- `OffsetAdmissible.period_dvd`;
- `OffsetAdmissible.all_add_mul_iff_period_dvd`;
- `OffsetAdmissible.least_positive_period`;
- `fixed_d_progression` and `fixed_d_progression_strictMono`;
- `family_I_positive`, `family_d2_24k5`, `family_d2_24k13`, and the four `family_d5_60k`· declarations.

For every explicit progression, the file also checks the instances $k = 0$ and $k = 1$. The theorem statement includes the positive denominators and the rational unit-fraction equality; the formalisation is therefore not merely a check of a polynomial identity.

8 Scope

The divisor split is classical, and offset choices of x belong to the standard toolkit for Egyptian-fraction problems. The contribution of this note is a compact theorem interface, an end-to-end Lean proof of the positive construction, an exact transport-step criterion and least-period theorem, and a small collection of formally checked infinite specialisations. The full Erdős–Straus conjecture remains outside its scope.

References

[1] L. J. Mordell, *Diophantine Equations*, Academic Press, 1969.

[2] R. C. Vaughan, *On a problem of Erdős, Straus and Schinzel*, Mathematika 17 (1970), 193–198, doi:10.1112/S0025579300002886.

[3] The Mathlib Community, *Mathlib: the mathematical library of Lean 4*, project repository.